

Understanding Employee Turnover Through HR Data Analytics

Fatema Shaker
DSB3

Problem Statement

Hr. Department has reported 50% of employees leaving, over the past 6 years, leading to company shrinkage.

Objectives

- **Track the workforce trend.** Compare hires and exits over the past six years and measure how exit changes each year, generally and department specifically.
- **Profile the exits.** Quantify how much each termination types represent relative to total employees and identify the job titles with the highest leaving rates
- **Pinpoint where people leave and why it might happen.** I identify the departments and divisions with the highest leaving numbers and %, and the relation to satisfaction rates
- **Measure time to exit.** Determine how long employees typically stay before leaving.

Key Metrics

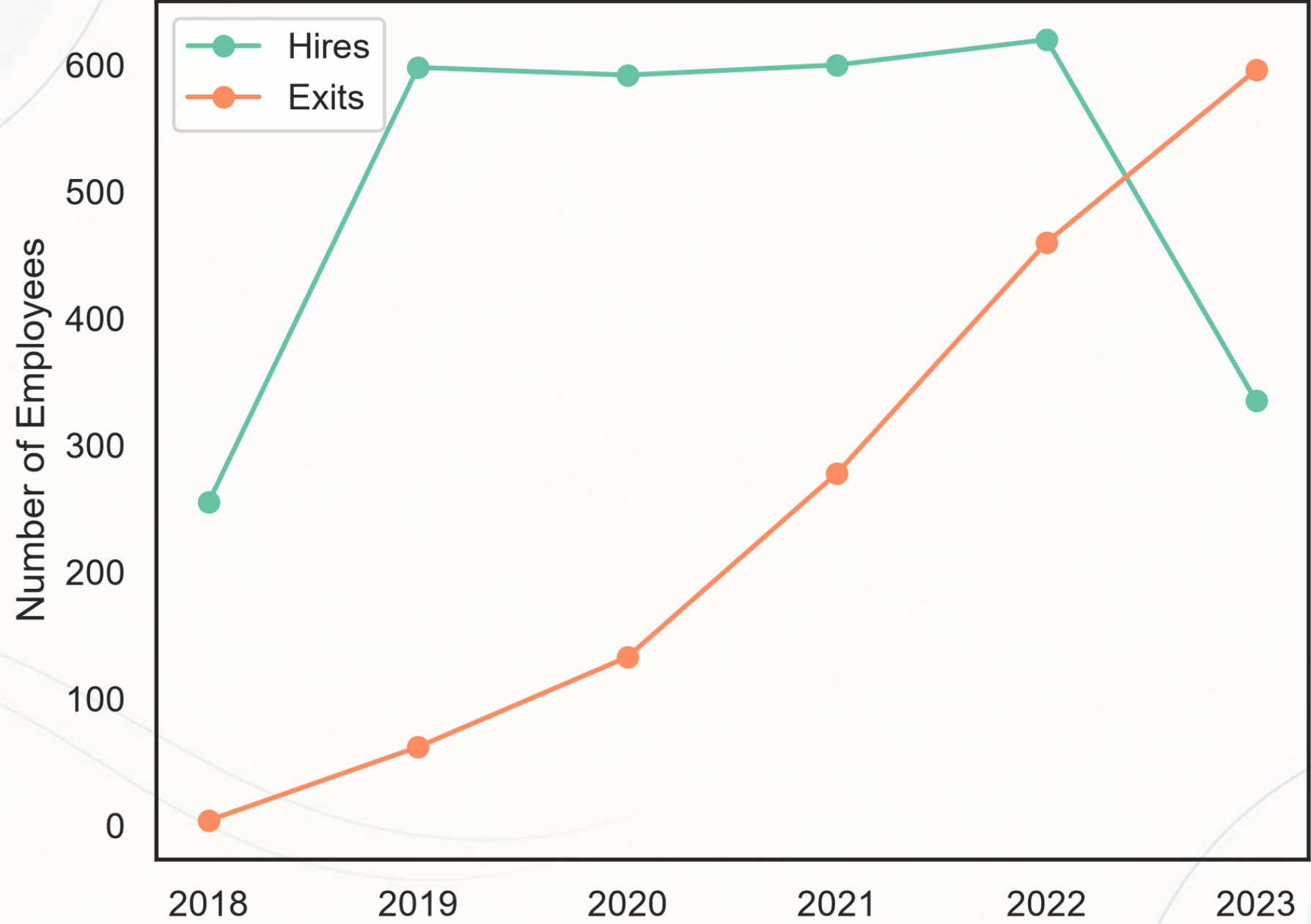
Total Exits
1467

Total Hires
3000

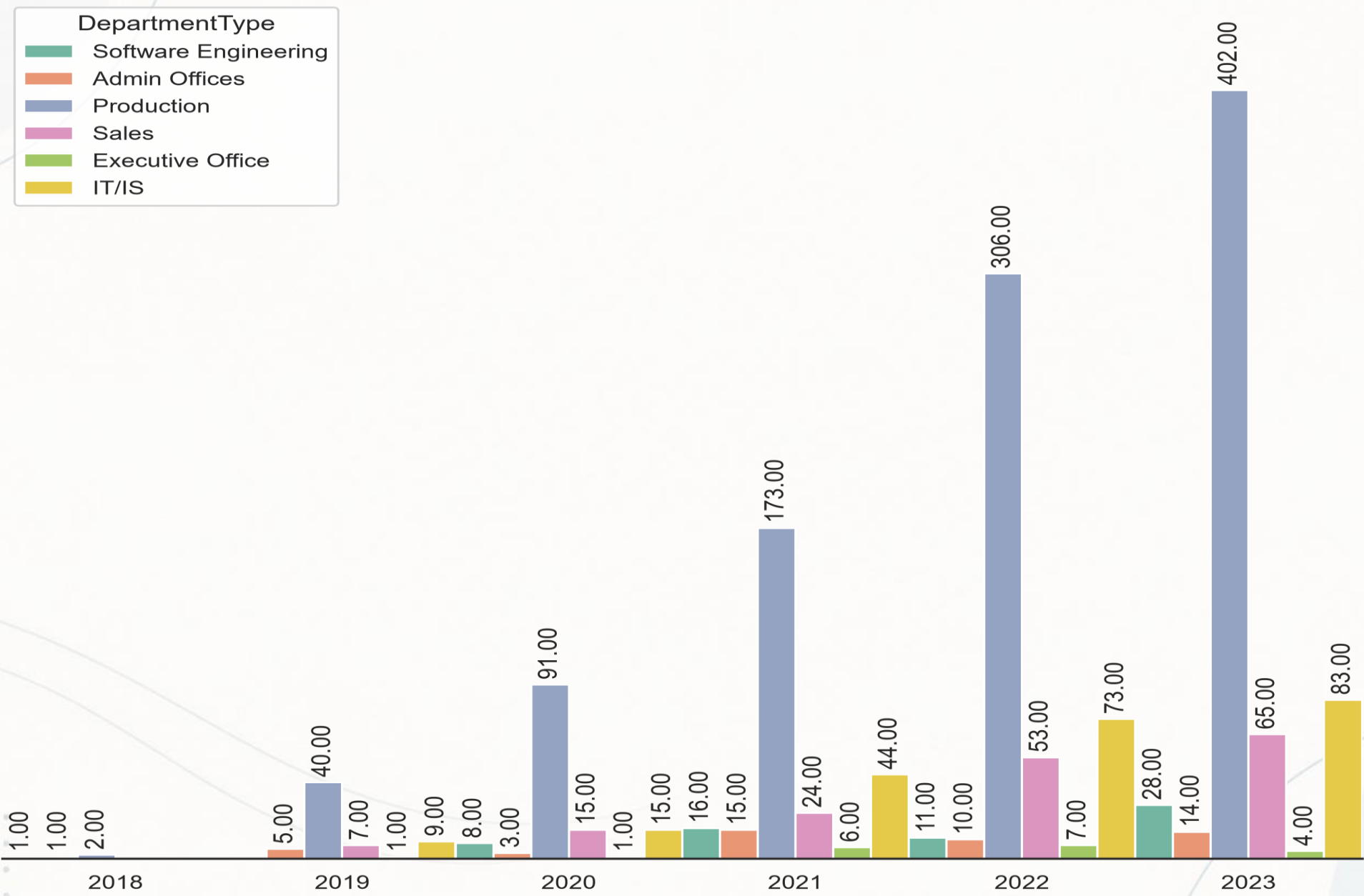
Period
2018-2023

Median Exit Years
1.1

Hires vs Exits by Year



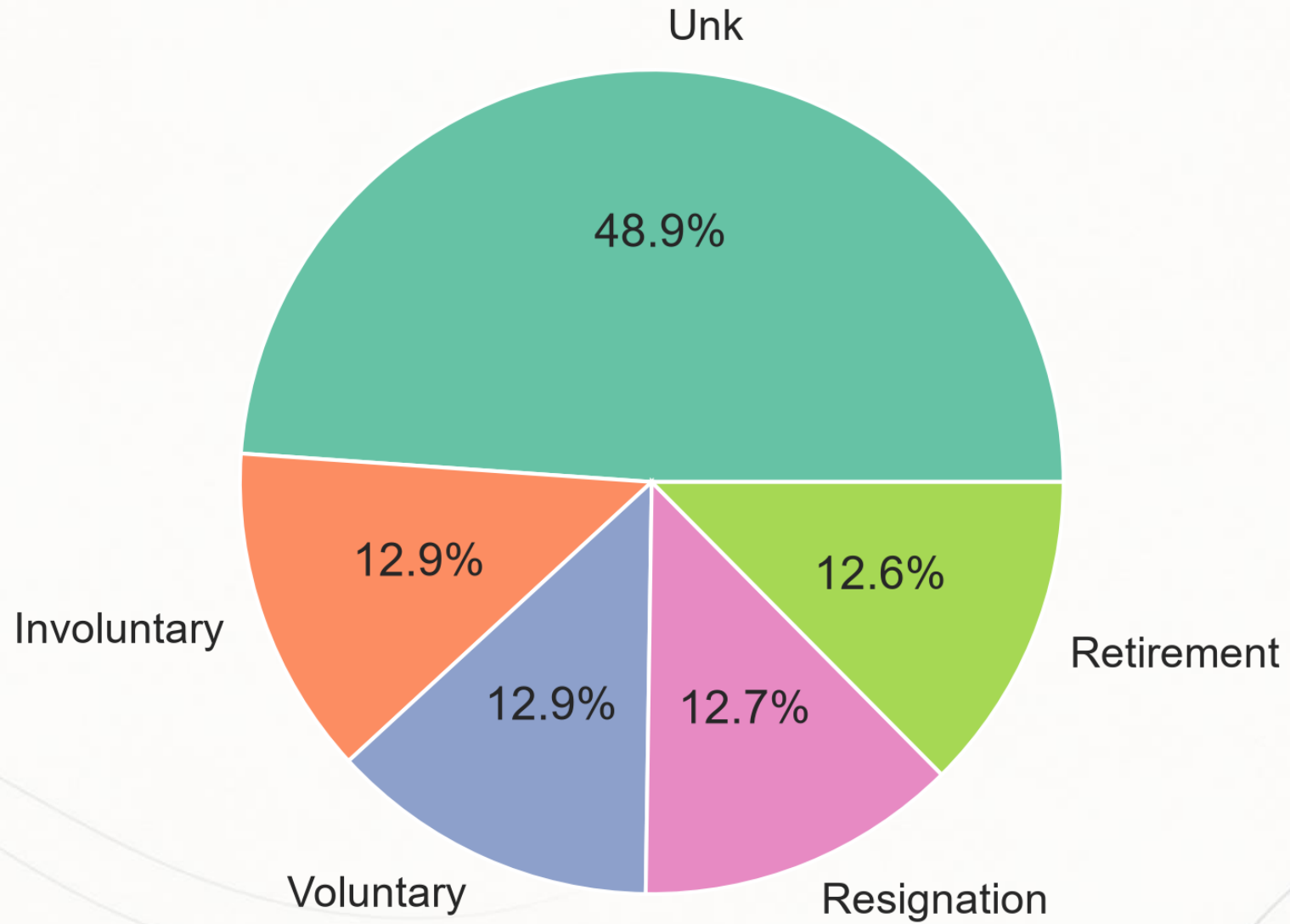
Department Exit Count per Year



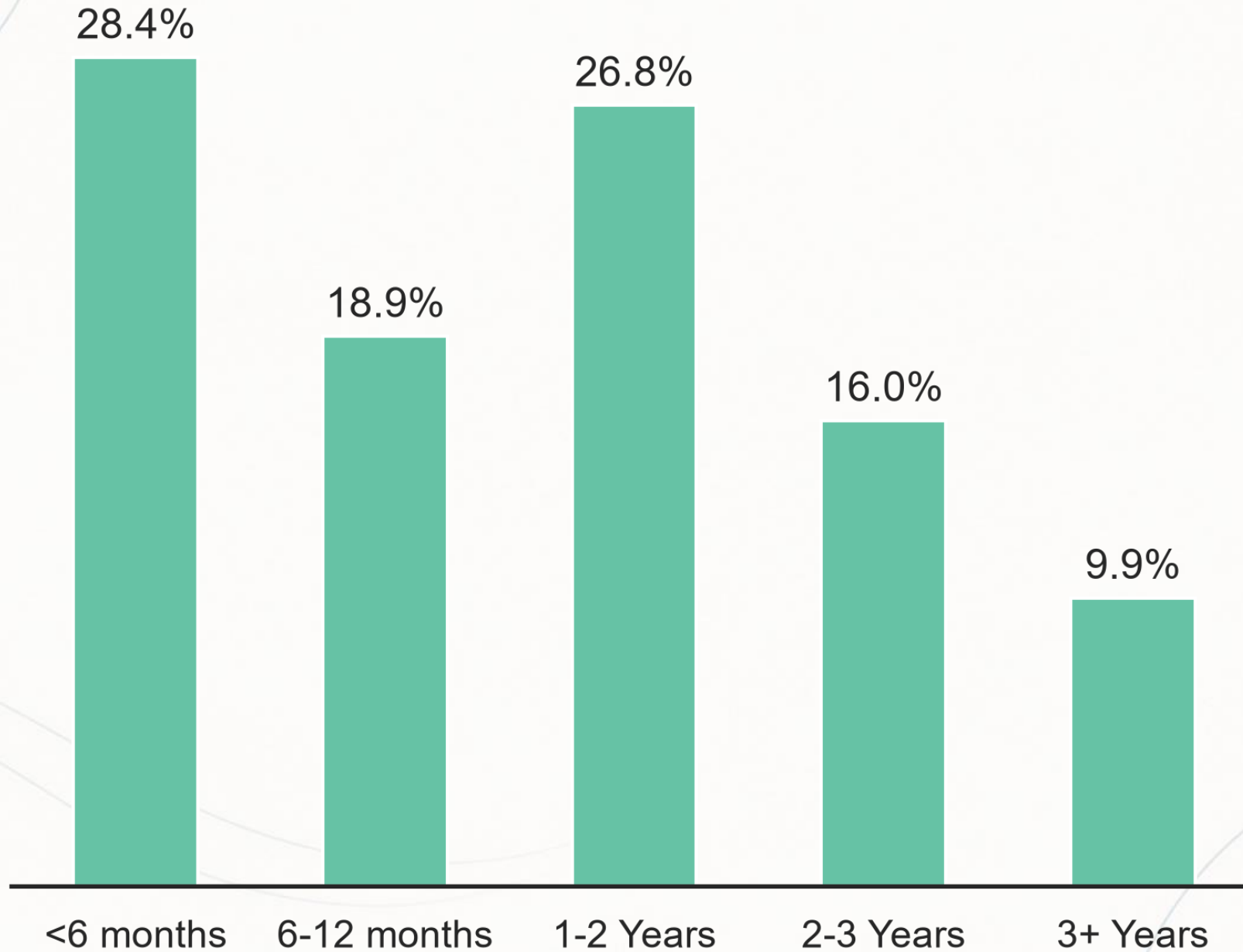
Department Exit Rate per Year

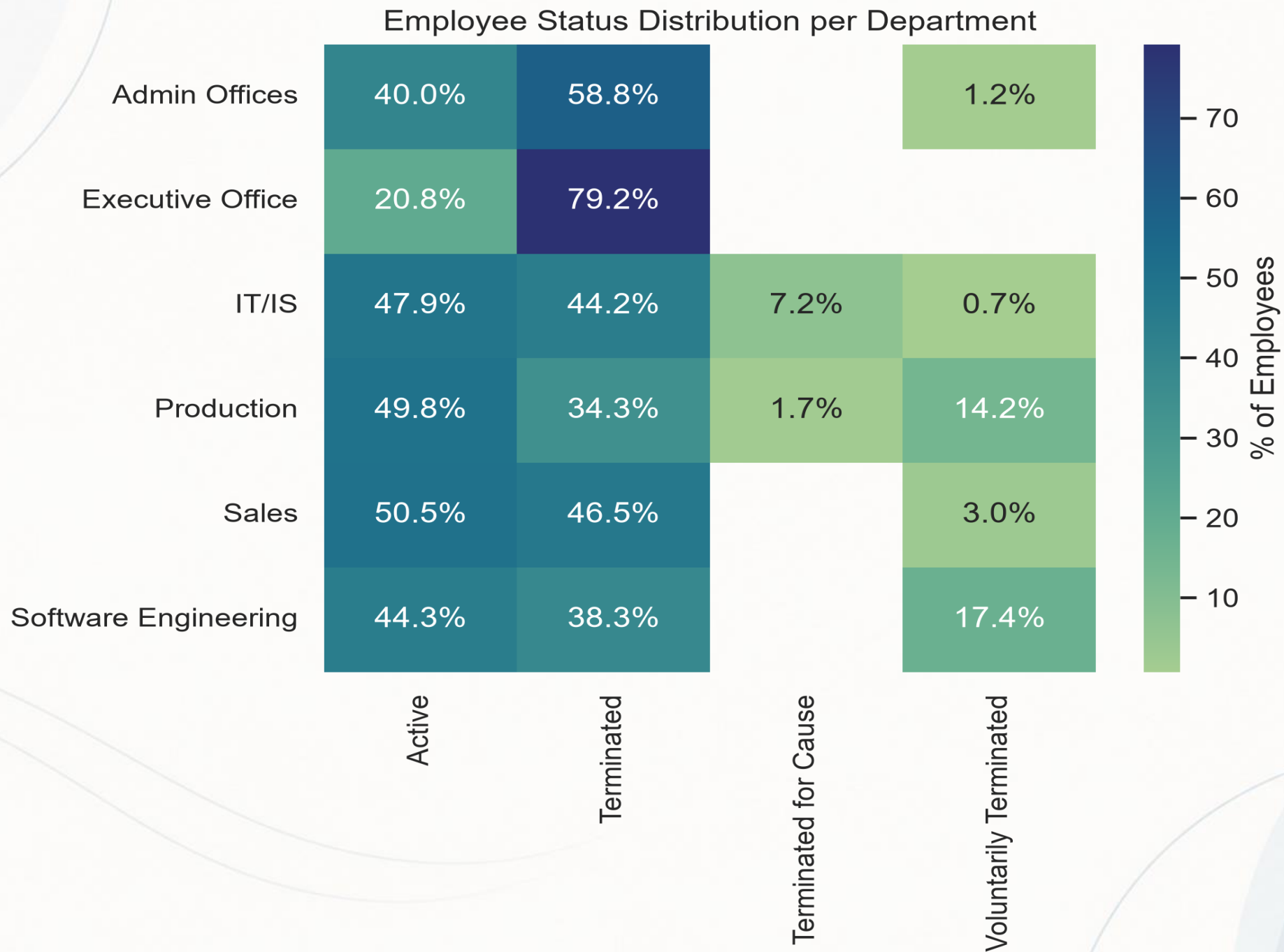


Termination Types Proportions

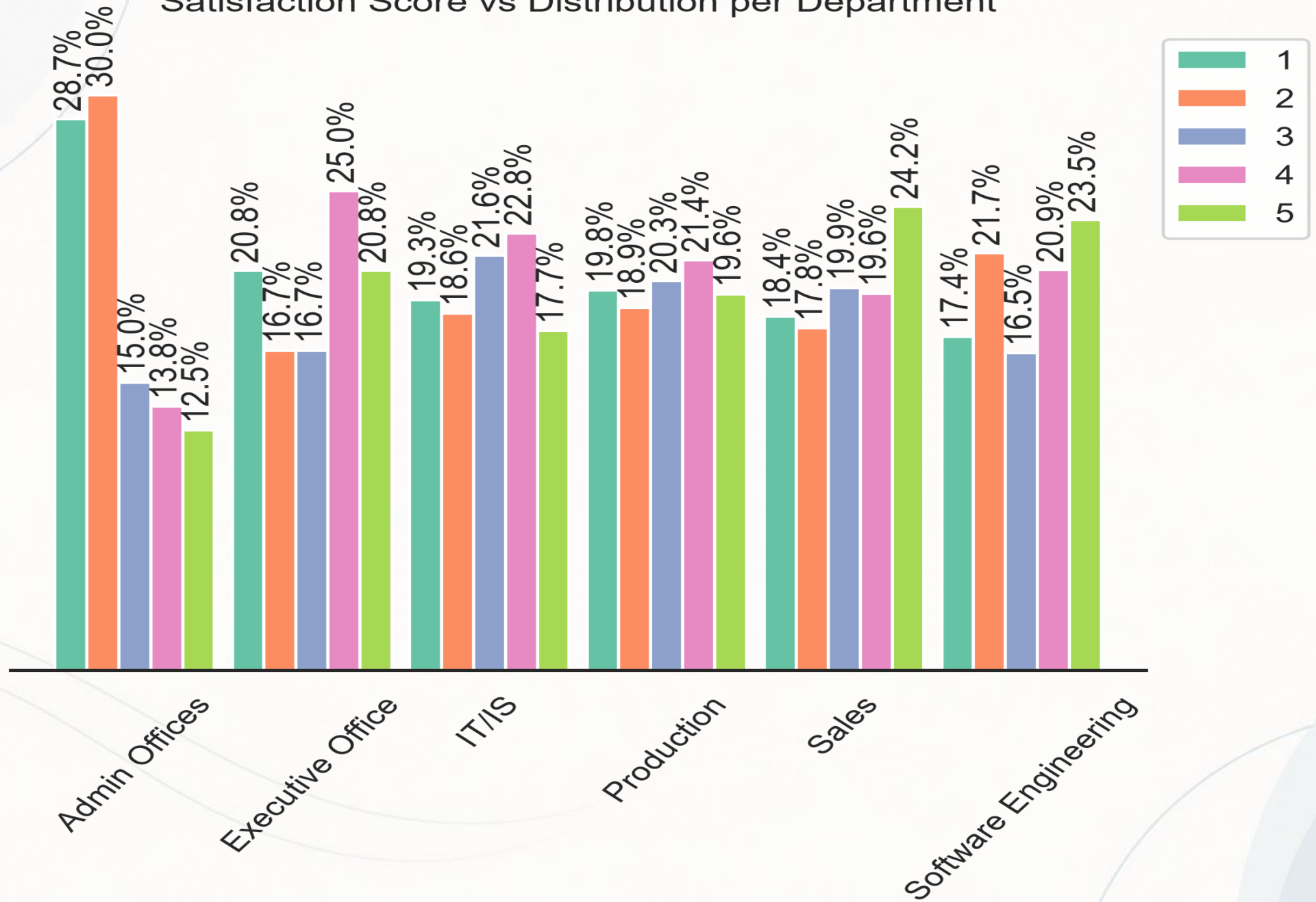


Employees% per Tenure

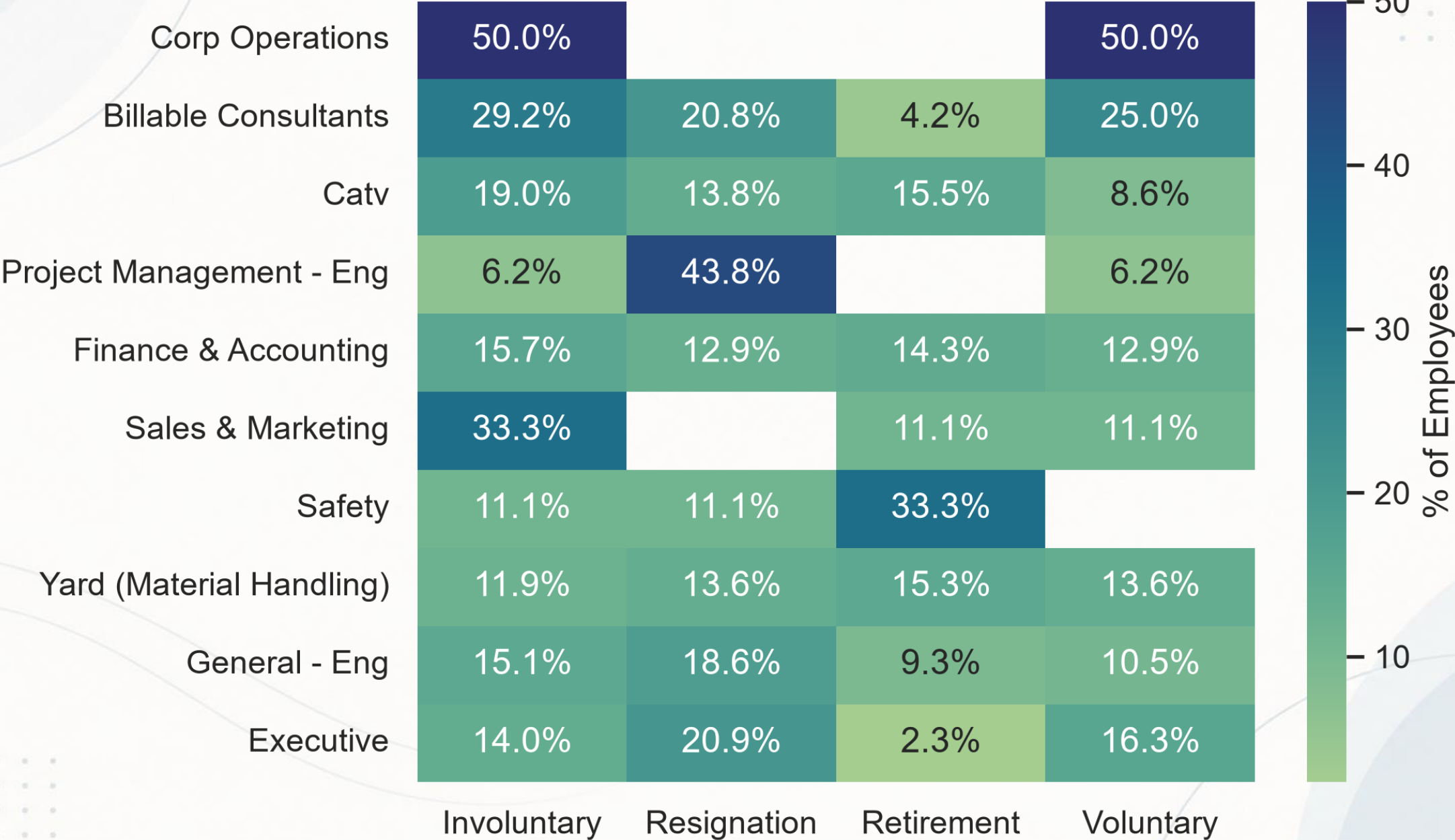




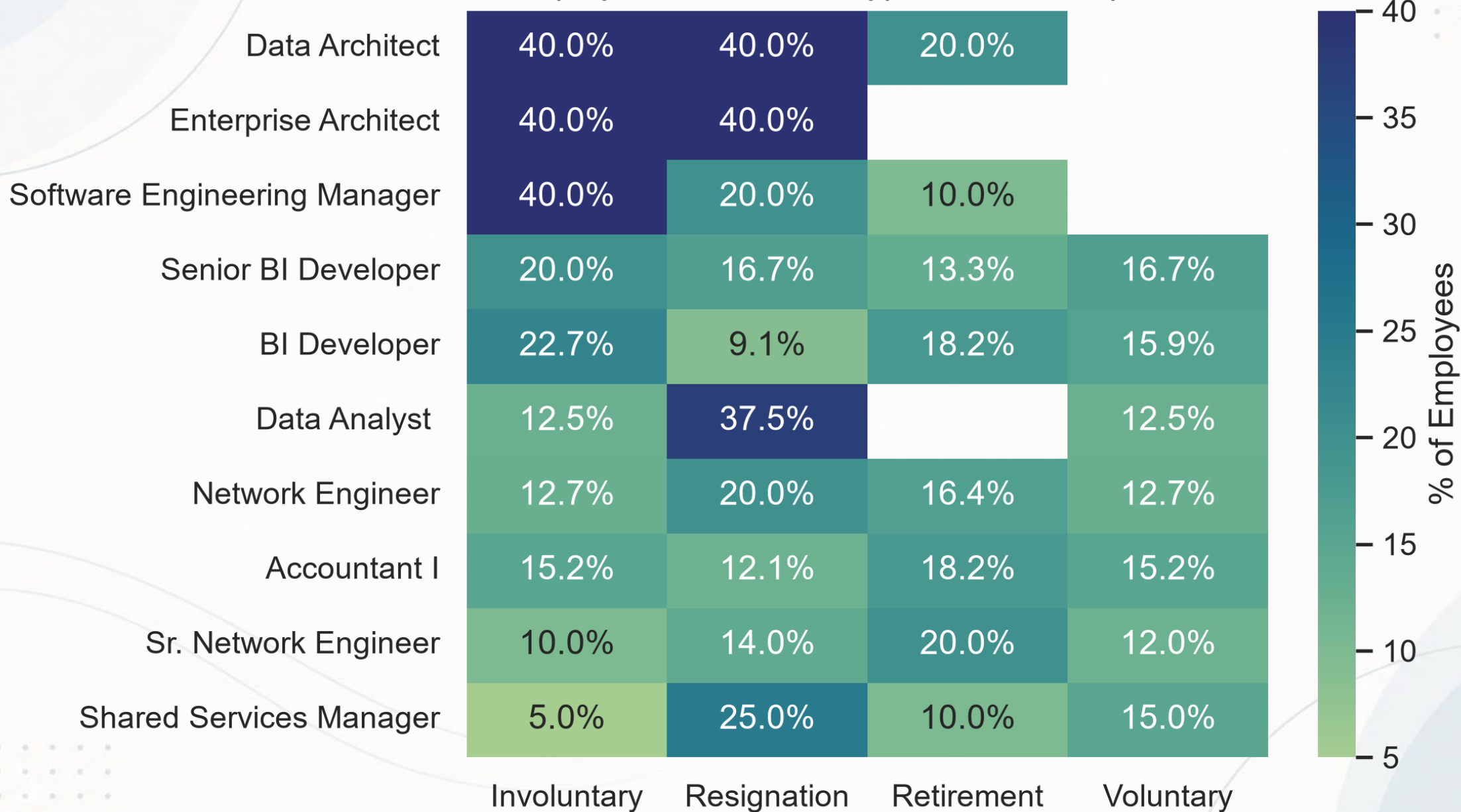
Satisfaction Score vs Distribution per Department



Employee Termination Type Distribution per Devision



Employee Termination Type Distribution per Title



Conclusion

- ❑ **Nearly half of the employees left the company between 2018 and 2023**
 - ❑ **Exit types: Involuntary, voluntary, resignation and retirement each account for about quarter of exits, so no single type drive the problem**
- ❑ **Timing: people leave early. The median stay before leaving is about 1.1 and among leavers 74% left within two years**
- ❑ **Departments: The executive Office and Admin had the highest leaving rates and peaked in 2021-2022, production lost the largest amount almost half of the exits in 2023**
 - ❑ **Satisfaction: Admin Offices reported the lower satisfaction.**
- ❑ **Data and architecture roles and software engineering managers show high exit rates**
- ❑ **The company can't explain the termination causes due to problems in termination description**
 - ❑ **Overall: Leaving is broad, early in employees tenure and growing.**

Recommendation

- ☐ **Write proper termination description to investigate the exit reasons accurately.**
- ☐ **Production employees contributes with nearly half of the losses in 2023, which worth further analysis.**
- ☐ **Admin offices and Executive low satisfaction rates indicates management or role problem leading to employees leaving the company.**
- ☐ **Monitor the IT roles (Data Architect, Enterprise Architect, Data Analyst) to lower the leaving rate.**
- ☐ **Do not hire the same number of people every year, the hiring process should be more flexible.**

Next Steps

Build a predictive model for leaving risk

Data Quality: Keep employee status in sync with exit dates, and validate survey and training dates against start and exit dates

Get real data, not AI generated data